

Block Randomisation

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Introduction

Block randomisation allocates clinical-trial participants in random permutations within fixed-size blocks, guaranteeing equal numbers of subjects in each treatment arm at every block boundary. It is the standard alternative to simple randomisation in clinical trials because it prevents the substantial arm-size imbalance that simple randomisation can produce in small trials, in early enrolment phases, or at any moment when the trial is paused for an interim analysis. Block randomisation is now mandated or strongly recommended by the ICH-E9 statistical-principles guideline, by CONSORT for randomisation reporting, and by virtually every regulatory authority's clinical-trial guidance.

Prerequisites

A working understanding of randomisation as the foundation of causal inference in trials, allocation concealment as the procedural safeguard against selection bias, and the difference between simple, blocked, and stratified randomisation.

Theory

With block size B and two equally-allocated arms, each block contains $B/2$ assignments of each arm in random order. At every block boundary the arm counts are exactly balanced; between boundaries the maximum imbalance is $B/2$. With fixed block size, an investigator who knows the block size and observes $B - 1$ allocations within a block can predict the final allocation — a serious threat in open-label trials. **Variable block sizes** (mixing, e.g., blocks of 4 and 6) defeat this predictability while preserving the boundary-balance guarantee.

For multi-arm trials with k arms in equal allocation, blocks must be multiples of k ; for unequal allocation (e.g., 2 : 1), blocks are multiples of the sum of allocation ratios.

Assumptions

Allocation concealment is preserved (the randomisation list is prepared in advance, kept off-site, and never available to enrolling investigators), the block-size distribution is documented in the statistical analysis plan but withheld from those who could exploit it, and randomisation is implemented through an interactive web-response system (IWRS) or equivalent with audit trail.

R Implementation

```
library(blockrand)

set.seed(2026)

fix <- blockrand(n = 40, num.levels = 2,
                 levels = c("A", "B"),
                 block.sizes = 2)
table(fix$treatment)
head(fix, 8)
```

```
var_b <- blockrand(n = 40, num.levels = 2,
                  levels = c("A", "B"),
                  block.sizes = c(2, 3))
head(var_b, 10)
```

Output & Results

`blockrand()` returns an allocation schedule with exactly equal arm counts across the requested n participants. With variable block sizes, the schedule blends blocks of different lengths in random order, preventing investigators from predicting the next allocation late in any single block. The schedule is typically exported to an IWRS and made available only to the trial pharmacist or unblinded statistician.

Interpretation

A reporting sentence: “Treatment allocation used block randomisation with variable block sizes of 4 and 6, generated by `blockrand` and managed via the trial’s interactive web-response system. Allocation was stratified by site and disease severity, with blocks nested within strata. The block-size distribution was documented in the SAP and concealed from enrolling investigators throughout the trial. Final arm counts were balanced (200 in each arm of the 400-patient trial).” Always report block-size distribution and concealment procedure.

Practical Tips

- Avoid fixed block size alone in open-label trials; variable block sizes are now the de facto standard for randomised clinical trials and are explicitly recommended by ICH-E9 because they prevent end-of-block predictability without sacrificing balance.
- Document the block-size distribution in the SAP but withhold the actual block sequence from enrolling investigators; sharing the block sequence (even informally) compromises allocation concealment and is a recurring cause of CONSORT-cited methodological flaws.
- For stratified designs (by site, disease severity, age category), nest blocks within strata so that each stratum maintains independent arm balance; this is standard practice in multicentre trials and prevents centre-by-treatment confounding.
- Very large blocks reduce guessability further but also relax the boundary-balance guarantee at any given moment in enrolment; small to moderate blocks (2 to 6 in two-arm trials) are the standard compromise and adequately balance most trials.
- Commercial randomisation services (IWRS / IRT) manage the list, preserve concealment, and provide a tamper-proof audit trail; for sponsor-led trials the cost is justified by the regulatory protection.
- For trials with more than two arms, use stratified blocked randomisation with appropriately sized blocks; permuted-block randomisation extends naturally to any number of arms with equal or unequal allocation ratios.

R Packages Used

`blockrand` for canonical fixed and variable-block randomisation with built-in stratification support; `randomizr` for tidyverse-friendly randomisation including blocked and stratified designs; `bcrm` for biased-coin and minimisation alternatives; `ldhmm` and `psborrow` for adaptive randomisation in more complex designs; `Mediana` for trial-design simulation including randomisation strategies.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren’t.**

- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Diagnostic testing: Se, Sp, PPV, NPV, LR
- Kappa, ICC, Bland–Altman
- Biomarker statistics (Youden, NRI, decision curves)
- TRIPOD-AI, fairness auditing, reproducibility at scale